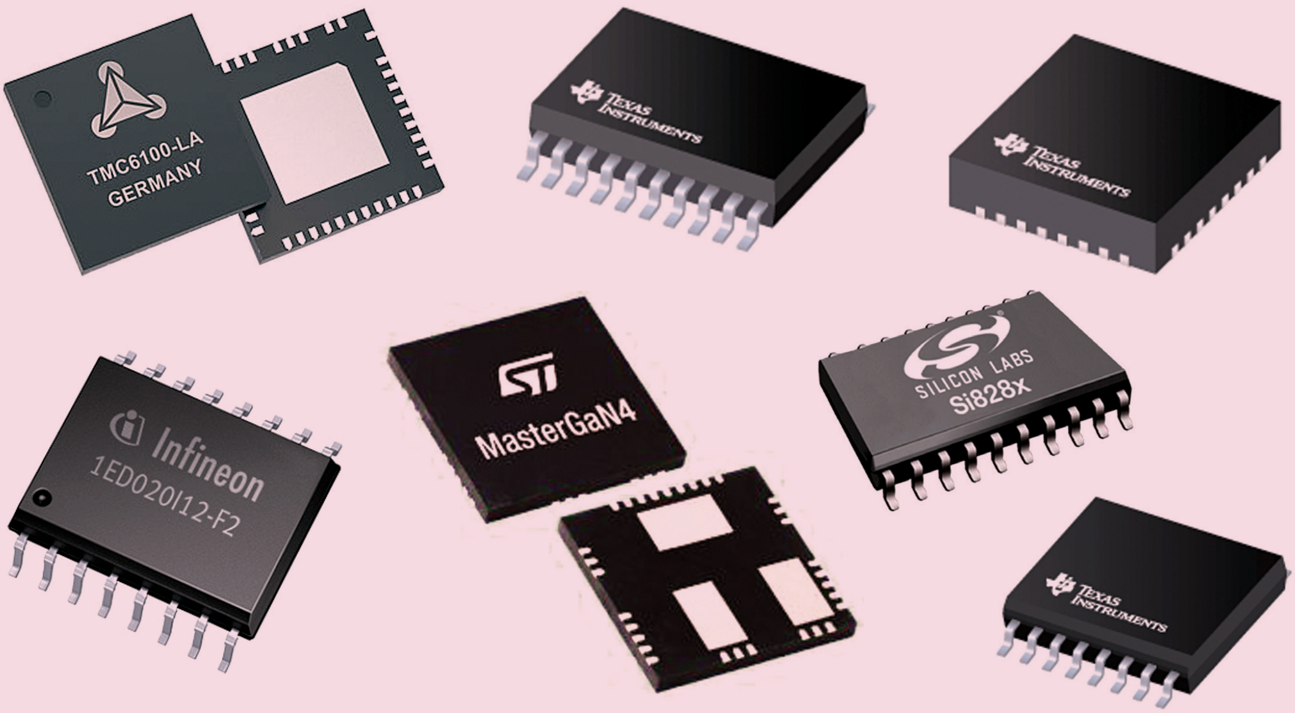
Emerging Trends In Gate Driver ICs

Find out the latest advancements in the gate driver ICs and how these technologies help develop more reliable power converters



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Gate drivers are an integral part of switching power converters as they provide an interface between microcontroller and power switches like MOSFETs and IGBTs. The gate driver cores usually include all commonly required driver functions for easy integration. And the gate drivers must perform reliably as the output of a switching DC-DC converter mainly depends on the behavior of gate driver circuits. Following are some new advancements in gate drivers that happened last year

Low propagation delay

Propagation delay is an important aspect of gate drivers to be taken into consideration, especially for high switching frequency applications. Propagation delay refers to the delay between input value change and output value change. For faster response time, minimum propagation delay is desirable.

Moreover, unmatched delay within multiple channel gate drivers degrades performance. Therefore, delay matching is required for robustness and high reliability when using channels parallelly. The latest gate drivers feature very low propagation delays.

The latest Microchip gate drivers feature 27ns typical delay times. Some of the Infineon EiceDRIVER gate driver ICs present propagation delay of 30ns and combine delay matching features. Texas Instruments DRV8300 features typical 4ns propagation delay matching. Texas Instruments UCC27289 features propagation delay of 16ns. On Semiconductors NCP51810